

QRM2

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Week 4

1 Movie budgets and profits

There is another argument going on in the movie studio. Bob claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. Chantal disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie profits”.

Step a

Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$Profits_i = \beta_0 + \beta_1 Budget_i + \varepsilon_i.$$

Print a summary of your estimated model. [2 points]

```
# Load the dataset and calculate profit in dollars
movies <- read.delim("movies5.tsv")
movies$profit <- movies$revenue - movies$budget

m1 <- lm(profit ~ budget, data = movies)
summary(m1)
##
## Call:
## lm(formula = profit ~ budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -387813422 -39322110   610960   8901755  950609114
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.307e+06  4.636e+06  -0.713    0.476
## budget       1.830e+00  8.798e-02  20.802   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 109900000 on 907 degrees of freedom
## Multiple R-squared:  0.323, Adjusted R-squared:  0.3223
## F-statistic: 432.7 on 1 and 907 DF, p-value: < 2.2e-16
```

Your Answer:

We estimated a linear regression with profit as the dependent variable and budget as the independent variable. Both are measured in dollars. The estimated model is: predicted profit = -3,307,162.73 + 1.8301 times budget. The model uses all 909 movies.

Step b

What is the estimated value of β_1 and how do you interpret it? [2 points]

```
coef(m1)["budget"]  
##    budget  
## 1.830122
```

Your Answer:

The estimated coefficient is 1.8301. A one-dollar higher budget is associated with approximately 1.83 dollars higher predicted profit, on average. Profit already equals revenue minus budget. This relationship does not prove that spending an extra dollar causes profit to increase by this amount.

Step c

Test for the null hypothesis that $\beta_1 \geq 0$. Report the p-value and state your conclusion. [2 points]

```
# Use the lower tail because the alternative is beta_1 < 0  
t_value <- summary(m1)$coefficients["budget", "t value"]  
p_value <- pt(t_value, df = df.residual(m1))  
p_value  
## [1] 1
```

Your Answer:

The null hypothesis is $H_0: \beta_1 \geq 0$, and the alternative is $H_1: \beta_1 < 0$. We use a 5% significance level. The t-value is approximately 20.80, and the one-sided p-value is approximately 1. We therefore do not reject H_0 . There is no evidence of a negative relationship. This test does not establish that β_1 is greater than 1.

Step d

Next, estimate the model:

$$\log(\text{Profits}_i) = \beta_0 + \beta_1 \log(\text{Budget}_i) + \varepsilon_i.$$

When creating the variables Log Profits and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. [2 points]

```
# Leave zero revenue or budget as NA in both log variables
movies$log_profit <- NA
movies$log_budget <- NA
valid_budget <- movies$budget > 0 & movies$revenue > 0
movies$log_budget[valid_budget] <- log(movies$budget[valid_budget])

# Log profit also requires a positive profit
valid_profit <- valid_budget & movies$profit > 0
movies$log_profit[valid_profit] <- log(movies$profit[valid_profit])

m2 <- lm(log_profit ~ log_budget, data = movies)
summary(m2)
##
## Call:
## lm(formula = log_profit ~ log_budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2956 -0.6776  0.1751  0.8725  3.1581
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.64769     0.76937   9.94  <2e-16 ***
## log_budget   0.58680     0.04482  13.09  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.346 on 461 degrees of freedom
## (446 observations deleted due to missingness)
## Multiple R-squared:  0.271, Adjusted R-squared:  0.2695
## F-statistic: 171.4 on 1 and 461 DF, p-value: < 2.2e-16
```

Your Answer:

We used natural logarithms. Both log variables are NA when revenue or budget is zero. Log profit is also NA when profit is zero or negative, because these values do not have a finite real logarithm.

The estimated intercept is 7.64769 and the coefficient of log budget is 0.58680. The model uses 463 movies with positive profit, revenue and budget. Its R-squared is 0.271, so it explains approximately 27.1% of the variation in log profit among these movies.

Step e

What is the estimated value of β_1 and how do you interpret it? [2 points]

```
coef(m2)["log_budget"]
## log_budget
## 0.5868046
```

Your Answer:

The estimated coefficient of log budget is 0.58680. Since this is a log-log model, a 1% higher budget is associated with approximately 0.59% higher profit. This relationship applies to the movies included in the model and does not prove that increasing a movie's budget causes its profit to increase.

Step f

Which model has better fit? The level-level model or the log-log model? Explain. [2 points]

```
c(level_level = summary(m1)$r.squared,  
  log_log = summary(m2)$r.squared)  
## level_level      log_log  
##    0.3230012    0.2710488
```

Your Answer:

The first model explains about 32.3% of the variation in profit. The second model explains about 27.1% of the variation in log profit.

The first model has a higher R-squared, but this does not necessarily mean it fits better. One model uses profit, while the other uses log profit. The first model also uses 909 movies, compared with 463 in the second model. Because of these differences, we cannot decide which model fits better just by comparing their R-squared values.

Step g

Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? [2 points]

```
# Check whether the 95% confidence interval is above 1  
confint(m1, "budget", level = 0.95)  
##           2.5 %    97.5 %  
## budget 1.65746 2.002784
```

Your Answer:

The level-level model supports Chantal's claim as a relationship: one dollar higher budget is associated with about 1.83 dollars higher profit. The usual 95% confidence interval for the coefficient is 1.66 to 2.00, entirely above 1. The log model also shows a positive relationship, but its coefficient measures percentage changes rather than dollars.

However, neither model proves that increasing a movie's budget causes higher profit. Genre, audience demand and other factors may affect both. We advise the studio to consider each movie's expected costs and revenue, rather than always choosing cheaper movies or assuming that every extra dollar will pay off.

2 First actor gender and movie budgets

Step a

Make a plot with a 95% confidence interval with the mean log of budget on the y-axis, and whether the first actor of the movie is male or female on the x-axis. What do you conclude? [2 points]

```
# For this question, use all positive budgets
movies$log_budget <- NA
positive_budget <- movies$budget > 0
movies$log_budget[positive_budget] <- log(movies$budget[positive_budget])

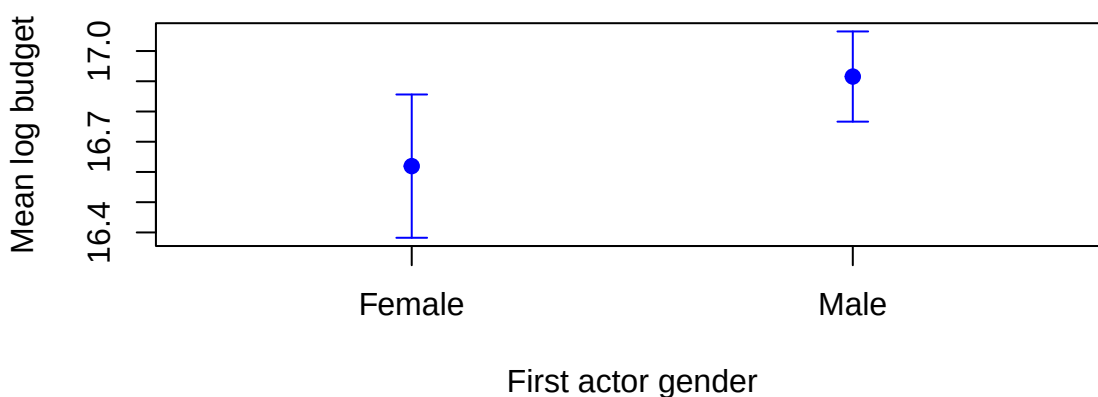
groups <- c("female", "male")
mean_log_budget <- numeric(2)
lower_budget <- numeric(2)
upper_budget <- numeric(2)

for (i in 1:2) {
  group_budget <- movies$log_budget[
    movies$first_actor_gender == groups[i]]
  group_budget <- group_budget[!is.na(group_budget)]
  mean_log_budget[i] <- mean(group_budget)
  interval <- t.test(group_budget, conf.level = 0.95)$conf.int
  lower_budget[i] <- interval[1]
  upper_budget[i] <- interval[2]
}

plot(1:2, mean_log_budget,
     main = "Mean log budget by first actor gender",
     xlab = "First actor gender", ylab = "Mean log budget",
     xaxt = "n", xlim = c(0.5, 2.5),
     ylim = range(lower_budget, upper_budget),
     pch = 19, col = "blue")
axis(1, at = 1:2, labels = c("Female", "Male"))

# Add the 95% confidence intervals
arrows(1:2, lower_budget, 1:2, upper_budget,
      angle = 90, code = 3, length = 0.08, col = "blue")
```

Mean log budget by first actor gender



Your Answer:

The mean log budget is 16.62 for the female group and 16.92 for the male group. The 95% confidence intervals are approximately 16.38 to 16.86 and 16.77 to 17.06. The male group has a higher mean log budget. The intervals overlap, but this alone does not tell us whether the difference is statistically significant.

For this separate question, we use all movies with a positive budget and a recorded first actor gender. We do not require positive revenue, because this model only concerns budget and gender.

Step b

Estimate the following simple OLS model:

$$\log(\text{budget}_i) = \beta_0 + \beta_1(\text{FirstActorMale})_i + \varepsilon_i.$$

Is the estimated coefficient for β_1 significantly different from zero? How do you interpret its estimate, and how does this relate to your conclusion in 2a? [2 points]

```
# Code male as 1 and female as 0; missing values remain NA
movies$first_actor_male <- as.numeric(movies$first_actor_gender == "male")
m3 <- lm(log_budget ~ first_actor_male, data = movies)
summary(m3)
##
## Call:
## lm(formula = log_budget ~ first_actor_male, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.9156  -0.5351   0.2872   0.9943   2.4945
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    16.6193     0.1280 129.884  <2e-16 ***
## first_actor_male  0.2963     0.1480   2.002   0.0457 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.707 on 702 degrees of freedom
## (205 observations deleted due to missingness)
## Multiple R-squared:  0.005674,    Adjusted R-squared:  0.004258
## F-statistic: 4.006 on 1 and 702 DF,  p-value: 0.04572

# Convert the log difference into a percentage difference
100 * (exp(coef(m3)["first_actor_male"]) - 1)
## first_actor_male
##           34.4858
```

Your Answer:

The coefficient is 0.2963, with a p-value of approximately 0.0457. We reject $H_0: \beta_1 = 0$ against $H_1: \beta_1$ is not zero at the 5% significance level. The model uses 704 movies: 178 in the female group and 526 in the male group. The male group has a mean log budget that is 0.2963 higher. This corresponds to a geometric mean budget about 34.49% higher. The direction agrees with the plot in step a. The regression finds a significant difference even though the separate confidence intervals overlap. It does not prove that the first actor's gender causes higher budgets.

Step c

Now, have a close look at your data frame. Can you find any instances of male first actors who are wrongly labelled as being female, or vice versa? What would such mislabelling mean for the coefficient you estimated under 2b? [2 points]

```
# Show two examples of incorrect labels in the original data
unique(movies[movies$first_actor %in% c("Morgan Freeman", "Jackie Chan"),
             c("first_actor", "first_actor_gender")])
##      first_actor first_actor_gender
## 33 Morgan Freeman          female
## 453 Jackie Chan          female
```

Your Answer:

Morgan Freeman and Jackie Chan are male actors but are labelled female in the dataset. Incorrect labels put movies in the wrong group and can bias the coefficient. If the mistakes are unrelated to budget within each true gender group, they generally make the estimated difference smaller. Otherwise, the bias can go either way. The results above use the original labels, so we should check them before drawing strong conclusions.

Week 5

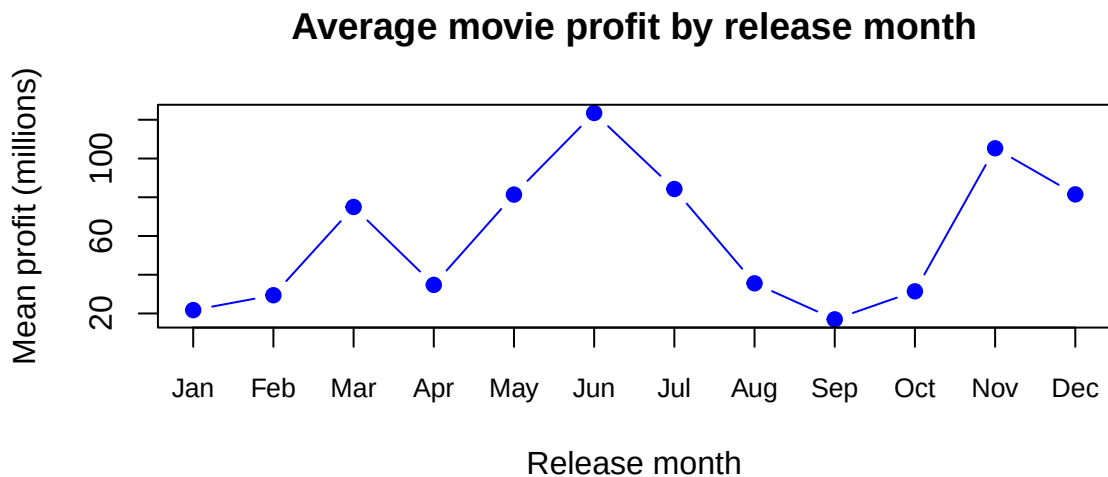
Movie release month and revenue predictions

Step a

Create a plot of the mean profits by month of release. Do you see any indication that month of release matters to the profits of the movie? [2 points]

```
# Calculate the mean profit for each release month
profits_by_month <- aggregate(profit ~ release_month,
                              data = movies, FUN = mean)

plot(profits_by_month$release_month,
     profits_by_month$profit / 1000000,
     type = "b", pch = 19, col = "blue", xaxt = "n",
     main = "Average movie profit by release month",
     xlab = "Release month", ylab = "Mean profit (millions)")
axis(1, at = 1:12, labels = month.abb, cex.axis = 0.8)
```



```
profits_by_month[which.max(profits_by_month$profit), ]
##   release_month   profit
## 6              6 123486518
```

Your Answer:

Mean profit differs between release months. June has the highest mean profit, at about 123.49 million, followed by November at 105.28 million. September has the lowest mean profit, at about 16.98 million.

The pattern goes up and down rather than increasing steadily throughout the year. This suggests that release month may matter, but the plot alone does not prove this. Movies released in different months may also differ in budget, genre and other characteristics.

Step b

Estimate an OLS model which has as dependent variable the log of profits of a movie, and as independent variables the log of budget, a dummy for whether the movie was released in English or not, and a linear term for the month of release. Show a summary of the resulting model and interpret each coefficient. [4 points]

```
# Code English as 1 and other languages as 0
movies$english <- as.numeric(movies$original_language == "en")

# Use the same complete sample for the models in steps b to e
model_data <- movies[complete.cases(movies[,
  c("log_profit", "log_budget", "english", "release_month"))], ]

m5b <- lm(log_profit ~ log_budget + english + release_month,
  data = model_data)
summary(m5b)
##
## Call:
## lm(formula = log_profit ~ log_budget + english + release_month,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2036 -0.6992  0.1714  0.8886  2.9585
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   6.96828    0.81336   8.567  <2e-16 ***
## log_budget     0.57697    0.04488  12.857  <2e-16 ***
## english        0.72135    0.34411   2.096   0.0366 *
## release_month  0.02106    0.01836   1.147   0.2519
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.34 on 459 degrees of freedom
## Multiple R-squared:  0.2807, Adjusted R-squared:  0.276
## F-statistic: 59.72 on 3 and 459 DF,  p-value: < 2.2e-16
```

Your Answer:

The model uses 463 movies with positive profit and budget. Each coefficient is interpreted while holding the other variables constant.

The intercept is 6.9683. It is the predicted log profit for a non-English movie with a one-dollar budget and release month zero. Since month zero does not exist, this has no useful practical interpretation.

The log budget coefficient is 0.5770. A 1% higher budget is associated with approximately 0.58% higher profit.

The English coefficient is 0.7214. English movies have a predicted log profit that is 0.7214 higher than non-English movies. This corresponds to about 105.72% higher geometric mean profit, using 100 times ($\exp(0.7214) - 1$), with the unrounded coefficient.

The release month coefficient is 0.0211. Moving one month later in the same calendar year is associated with approximately 2.13% higher profit. These relationships do not prove that changing budget, language or release month causes profit to change.

Step c

Test for the hypothesis that the coefficient that belongs to month of release is zero. [2 points]

```
summary(m5b)$coefficients["release_month", ]
##      Estimate Std. Error    t value    Pr(>|t|)
## 0.02106297 0.01836138 1.14713482 0.25192350
```

Your Answer:

The null hypothesis is $H_0: \beta_{\text{month}} = 0$, and the alternative is $H_1: \beta_{\text{month}} \neq 0$. The t-value is approximately 1.147 and the p-value is 0.2519. Since this is above 0.05, we do not reject H_0 . There is no statistically significant linear month trend in this model. This does not rule out a non-linear pattern across months.

Step d

Based on your plot in a, do you consider the choice that month of release enters the model linearly under b reasonable? Estimate a specification that allows for a more flexible curve. In this new specification, test for the null hypothesis that month of release does not impact profits. This might require testing multiple terms at once. [4 points]

```
# Treat month as a category, with January as the reference month
m5d <- lm(log_profit ~ log_budget + english + factor(release_month),
          data = model_data)
summary(m5d)$coefficients
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)    7.65331054  0.84503788   9.0567663 4.140274e-18
## log_budget     0.53453882  0.04786166  11.1684143 1.005551e-25
## english        0.79437367  0.34582313   2.2970518 2.207515e-02
## factor(release_month)2 -0.41090814  0.38084883  -1.0789271 2.811995e-01
## factor(release_month)3 -0.04083193  0.32203765  -0.1267924 8.991615e-01
## factor(release_month)4 -0.27005408  0.34353186  -0.7861107 4.322173e-01
## factor(release_month)5  0.49839240  0.32407370   1.5378983 1.247779e-01
## factor(release_month)6  0.59488897  0.31297945   1.9007285 5.797770e-02
## factor(release_month)7  0.15807347  0.31904346   0.4954606 6.205174e-01
## factor(release_month)8  0.15656912  0.30236677   0.5178119 6.048445e-01
## factor(release_month)9 -0.21889686  0.29299624  -0.7470978 4.553955e-01
## factor(release_month)10 0.09947628  0.29975639   0.3318571 7.401520e-01
## factor(release_month)11 0.43306818  0.31456575   1.3767175 1.692857e-01
## factor(release_month)12 0.16913548  0.29218396   0.5788664 5.629696e-01

# Jointly test all 11 month coefficients using the same movies
m5d_restricted <- lm(log_profit ~ log_budget + english,
                     data = model_data)
anova(m5d_restricted, m5d)
## Analysis of Variance Table
##
## Model 1: log_profit ~ log_budget + english
## Model 2: log_profit ~ log_budget + english + factor(release_month)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      460 826.10
## 2      449 792.83 11    33.267 1.7127 0.06788 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Your Answer:

A linear month effect does not describe the plot well because profit rises and falls across the year. We therefore treat month as a category. January is the reference month, and the other 11 months each have their own coefficient. This allows a separate pattern for each month.

The null hypothesis is that all 11 month coefficients equal zero. The alternative is that at least one is not zero. The joint F-test gives $F(11, 449) = 1.7127$ and $p = 0.0679$. At the 5% significance level, we do not reject H_0 . The evidence is not strong enough to conclude that the months jointly improve the model, after accounting for budget and language. Both models use the same 463 movies.

Step e

One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Based on your model under d, what would you advise the movie studio regarding the timing of the release of the movie? [2 points]

```
# Compare months while keeping budget and language the same
month_options <- data.frame(log_budget = mean(model_data$log_budget),
                             english = 1, release_month = 1:12)
predicted_log_profit <- predict(m5d, newdata = month_options)
month.name[which.max(predicted_log_profit)]
## [1] "June"
```

Your Answer:

June has the highest predicted log profit when budget and language are held constant. However, the joint test in step d is not significant at 5%, so we cannot confidently recommend June as the best month. The pattern is also based only on movies with positive profits and budgets.

The model predicts log profit, not revenue directly. For a fixed budget, higher profit also means higher revenue, but this does not prove that moving a release to June will increase revenue. We would consider June as one option while also looking at competition, audience demand and the type of movie.

Step f

The movie studio that you work at is releasing a new movie in 2026. It will be an English-spoken Thriller movie with a budget of 40,000,0000.

Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 99% prediction interval. [6 points]

```
# Predict revenue, including movies that did not make a profit
revenue_data <- subset(movies, revenue > 0 & budget > 0)
revenue_data$log_revenue <- log(revenue_data$revenue)
revenue_data$thriller <- as.numeric(revenue_data$genre == "Thriller")

m5f <- lm(log_revenue ~ log_budget + english + thriller + release_year,
          data = revenue_data)
summary(m5f)
##
## Call:
## lm(formula = log_revenue ~ log_budget + english + thriller +
##     release_year, data = revenue_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.8618  -0.6587   0.1927   0.7990   6.5821
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -30.28042   18.025919  -1.680   0.0935 .
## log_budget     0.878029    0.043411   20.226 <2e-16 ***
## english        0.361804    0.317094    1.141   0.2543
## thriller       0.029167    0.159035    0.183   0.8545
## release_year   0.016263    0.008977    1.812   0.0705 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.477 on 624 degrees of freedom
## Multiple R-squared:  0.4062, Adjusted R-squared:  0.4024
## F-statistic: 106.7 on 4 and 624 DF, p-value: < 2.2e-16

# Use a budget of 40 million, as in our original code
new_movie <- data.frame(log_budget = log(40000000), english = 1,
                        thriller = 1, release_year = 2026)

# Predict one movie and convert the result back to dollars
prediction_log <- predict(m5f, newdata = new_movie,
                         interval = "prediction", level = 0.99)
exp(prediction_log)
##           fit           lwr           upr
## 1 100779498 2111898 4809184565
```

Your Answer:

We interpret the budget written as “40,000,0000” as 40 million dollars, as in our original code. We estimated log revenue using log budget, English language, Thriller genre and release year. The model uses 629 movies with positive revenue and budget, including movies that made a loss.

For the English-spoken Thriller released in 2026, the back-transformed revenue prediction is approximately 100.78 million dollars. This is the model’s predicted median revenue, rather than its arithmetic mean. The 99% prediction interval for this individual movie is approximately 2.11 million to 4,809.18 million dollars. We used a prediction interval, not a confidence interval for the average revenue.

The interval is very wide, so the prediction is uncertain. It relies on the regression assumptions, including approximately normal errors with constant variance on the log scale. Our data cover 1990 to 2016, so using the model for 2026 also assumes that the estimated relationship continues beyond the observed years.